Bo Li | 李波

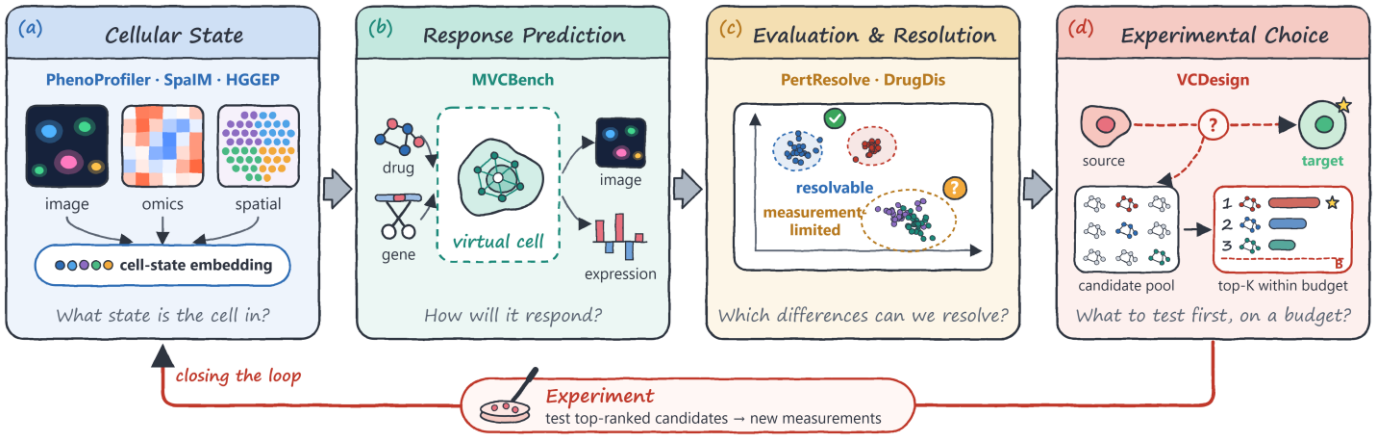
Ph.D. Candidate in Computer Science, University of Macau
Visiting Student, National University of Singapore

Multimodal Virtual Cells · **Perturbation Modeling** · **Intervention Design**

Email
Homepage
Google Scholar
GitHub
ORCID

RESEARCH PROFILE

I develop AI methods that turn multimodal cellular measurements into experimental decisions. My work follows one loop, shown below: learning representations of cellular state from microscopy, single-cell and spatial omics (two first-author Nature Communications papers); predicting perturbation responses across modalities and cellular contexts; testing whether predicted gains are reproducible at the resolution an assay can actually measure; and ranking which interventions to test next under a finite budget. The goal is decision-grade virtual cells, judged by the experiments they choose, not only by prediction accuracy.



EDUCATION & APPOINTMENTS

University of Macau — Aug 2024 – Jun 2027 (expected)
Ph.D. in Computer Science | Advisor: Prof. Bob Zhang; Co-advisor: Prof. Qianqian Song

National University of Singapore — Jun 2026 – Jun 2027
Visiting Ph.D. Student | Host: Prof. Yang Zhang

Beijing University of Technology — Sep 2021 – Jul 2024
M.Eng. in Electronic Information | Advisors: Prof. Yong Zhang and Prof. Baocai Yin

Beijing Information Science & Technology University — Sep 2017 – Jul 2021
B.Eng. in Robotics Engineering | Advisor: Prof. Hongbo Huang

9	2	569	1,000+
First-author journal articles	First-author *Nature Communications*	citations · h-index 13 Google Scholar · Oct 2026	**GitHub stars** Across open-source projects

RESEARCH PROGRAM & SELECTED PUBLICATIONS

7 core works · † equal contribution

CELLULAR STATE PhenoProfiler · SpaIM · HGGEP

What representations of cellular state can be learned from imaging and omics measurements?

1. **PhenoProfiler: Advancing phenotypic learning for image-based drug discovery**
   Bo Li, Bob Zhang, …, Mengran Li, Yong Zhang, Qianqian Song · *Nature Communications* 17, 793 (2026)
   Learns phenotypic profiles directly from high-content microscopy images, without hand-crafted feature-extraction pipelines, for compound profiling.
2. **SpaIM: Single-cell spatial transcriptomics imputation via style transfer**
   Bo Li†, Ziyang Tang†, …, Baijian Yang, Jing Su, Qianqian Song · *Nature Communications* 16, 7861 (2025)
   Treats imputation of unmeasured genes in spatial transcriptomics as style transfer from single-cell references.
3. **Gene expression prediction from histology images via hypergraph neural networks**
   Bo Li, Yong Zhang, …, Mengran Li, Guangyu Wang, Qianqian Song · *Briefings in Bioinformatics* 25(6), bbae500 (2024)
   Predicts spatial gene expression from H&E histology with hypergraphs that capture higher-order tissue structure.

RESPONSE PREDICTION MVCBench

How well do predicted perturbation responses transfer across modalities, compounds and cellular contexts?

4. **MVCBench: A multimodal benchmark for drug-induced virtual cell phenotypes**
   Bo Li†, Qing Wang†, Shihang Wang†, …, Xiaojun Yao, Chuxia Deng, Qianqian Song · *bioRxiv* (2026)
   Benchmarks image- and expression-based perturbation prediction under compound, cell-line, plate and dataset shifts.

EVALUATION & RESOLUTION PertResolve · DrugDis

When does a predicted gain reflect biology that the assay can reproducibly resolve?

5. **Measurement resolution constrains fine-grained perturbation prediction**
   Bo Li†, Chengyang Zhang†, Mengran Li†, …, Yuhao Yi, Jiancheng Lv, Yang Zhang · *Under review* (2026)
   Asks which fine-grained perturbation distinctions an assay can reproducibly resolve, separating model limits from measurement limits.

6. **Disentangling general and context-specific effects in drug-response prediction**
   Bo Li†, Chengliang Liu†, Yuzhong Peng†, …, Shenghui Huang, Chuxia Deng, Yang Zhang · *Under review* (2026)
   Decomposes drug-response prediction into general drug, general sample and drug–sample interaction effects, showing that aggregate accuracy can hide weak recovery of the context-specific part.

EXPERIMENTAL CHOICE VCDesign

Given a target cellular state and a finite budget, which interventions are worth testing next?

7. **VCDesign: Finite-budget intervention design for virtual cells**
   Bo Li†, Lin Wang†, Bob Zhang, …, Chengliang Liu, Zhiyuan Liu, Yang Zhang · *Under review* (2026)
   Ranks candidate interventions toward a target state under a fixed budget, scores the selected set against independently measured held-out outcomes, and transfers evidence to candidates with no measured response.

SELECTED OTHER FIRST-AUTHOR PUBLICATIONS

full list of 39 works on Google Scholar

8. **Multi-scale hypergraph-based feature alignment network for cell localization**
   Bo Li, Yong Zhang, Chengyang Zhang, Xinglin Piao, Yongli Hu, Baocai Yin · *Pattern Recognition* 149, 110260 (2024)
9. **Exponential distance transform maps for cell localization**
   Bo Li, Jie Chen, Hang Yi, …, Yongquan Yang, Qikui Zhu, Hong Bu · *Eng. Appl. Artif. Intell.* 132, 107948 (2024)
10. **Lite-UNet: A lightweight and efficient network for cell localization**
    Bo Li, Yong Zhang, Yunhan Ren, Chengyang Zhang, Baocai Yin · *Eng. Appl. Artif. Intell.* 129, 107634 (2024)

Co-corresponding-author manuscript

11. **When do single-cell response distributions improve drug retrieval?**
    Huihong Wang, Xiaoping Liu, Chen Chen, Qiuyan Sun#, **Bo Li#**, Bob Zhang# · *Under review* (2026)

OPEN SOURCE & RESEARCH SYSTEMS

1,000+ GitHub stars in total

Nature-Paper-Skills	Agent skills for scientific writing, revision, auditing and resubmission.	**583 stars**
Awesome Virtual Cell	Research landscape: 328 papers · 20 themes · interactive map.	**324 stars**
DrugClaw	Agentic RAG for drug research · 57 skills · 15 task categories.	**116 stars**

INVITED TALKS & MENTORING

Invited talk: PhenoProfiler, Prof. Guangyu Wang's group, Weill Cornell Medicine & Houston Methodist — Feb 2025

Mentoring: co-mentored four graduate students at Beijing University of Technology; they first-authored six papers (New Phytologist, IEEE JBHI, ACM TOMM, CVIU, two arXiv preprints).

HONORS, AWARDS & PATENTS

National Scholarship, China's top graduate award — 2022
Top 100 Graduates, BJUT — 2024
First-Class Academic Scholarship, BJUT — 2022, 2023
First Prize, S&T Innovation Scholarship, BISTU — 2021
Beijing Outstanding Graduate — 2021, 2024
Xiaomi Scholarship — 2023
Ph.D. Full Scholarship, UM — 2024

Patents: two granted Chinese invention patents on cell localization, co-inventor (2026): CN115457547B · CN115810046B

ACADEMIC SERVICE

Reviewer: ICLR 2027 (conference); Science Advances; IEEE TPAMI, TIP, TNNLS; Medical Image Analysis; Bioinformatics; Briefings in Bioinformatics; Pattern Recognition; BMC Biology; Eng. Appl. Artif. Intell.; Expert Systems with Applications (journals).